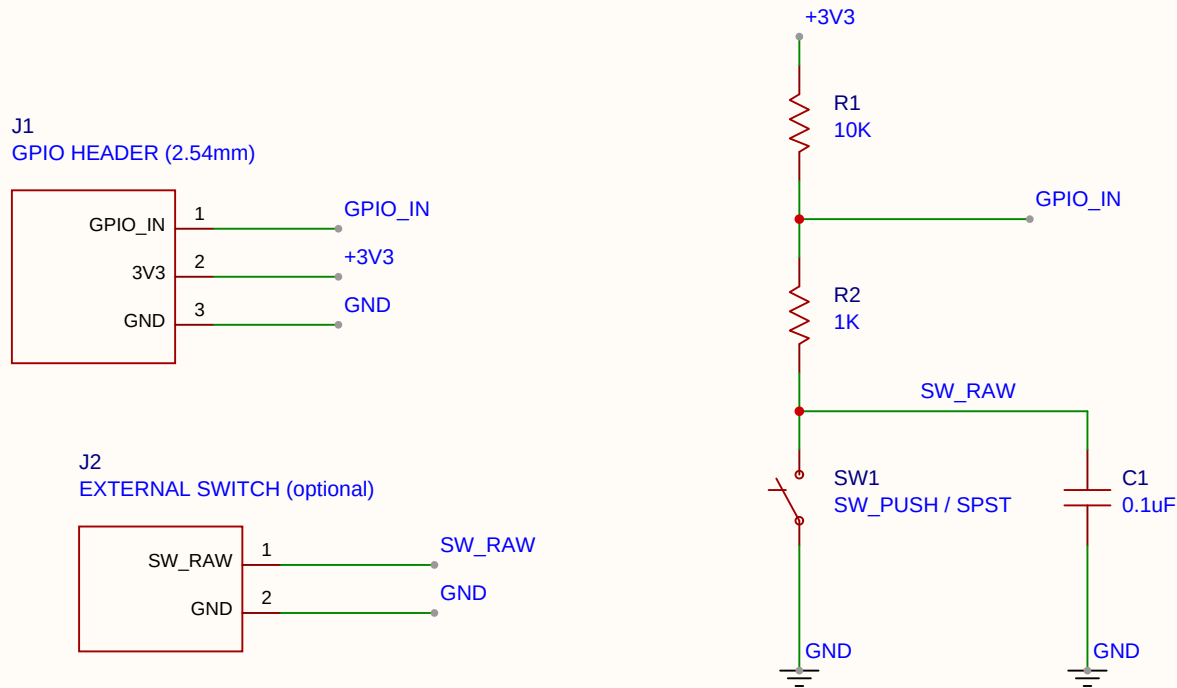


GPIO Push-Button or Switch Input

3.3 V logic | Editable schematic | Rev 1.1 | 2026-10-04



Operation: released / open = HIGH; pressed / closed = LOW.

R1 10K pull-up gives the idle level. R2 1K is in series with the switch; C1 0.1uF is across the switch.

Same network as Maker Pi RP2040 sheet 2 PUSH BUTTONS (R19 10K, R20 1K, S3, C26 0.1uF).

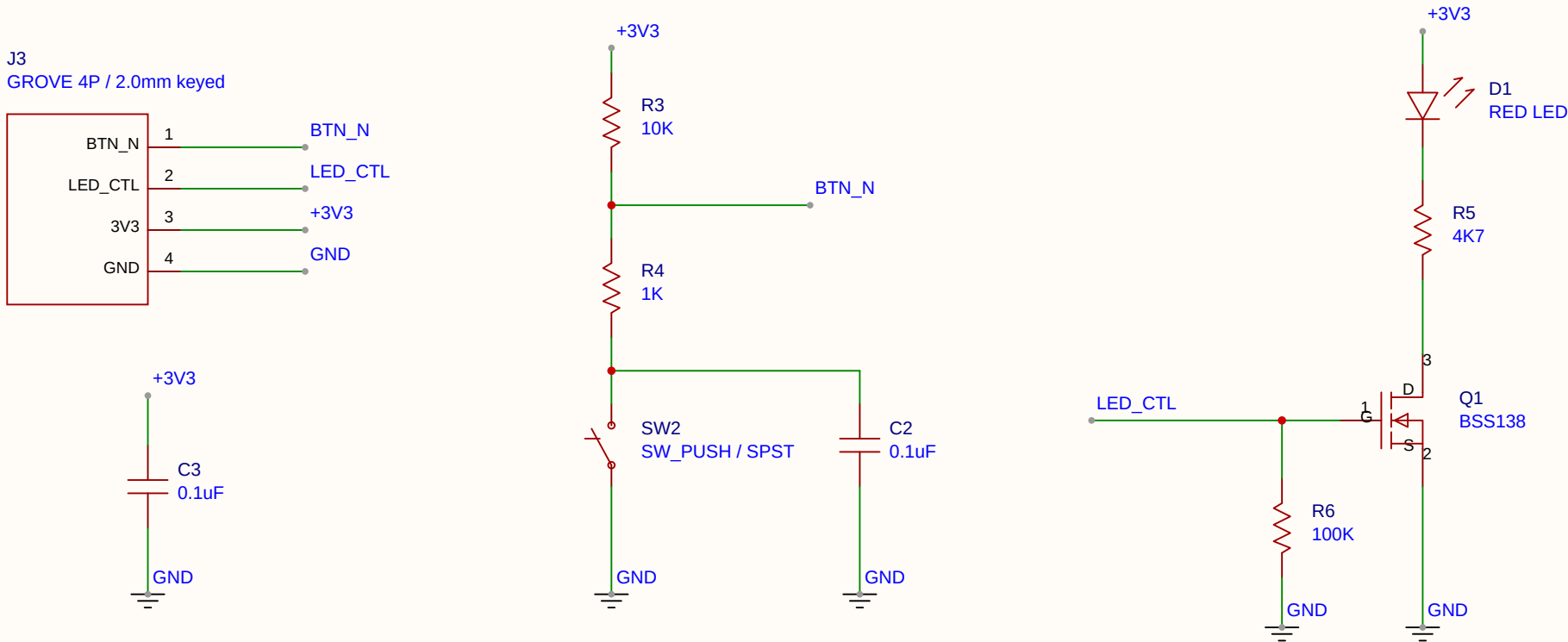
J2 accepts a dry-contact switch in parallel with SW1. Configure the MCU pin as an input.

References: Maker Pi RP2040 sheet 2; Maker Pi Pico sheet 1; ETT ET-ESP32 LAB V1 input switch block.

Added to the reference: J1 header, optional J2. 3.3 V only. SW1 uses switch pins 1 and 3; pads 2 and 4 unmapped - verify 1 and 3 are opposite contacts before wiring.

GROVE_4P_Button MOSFET_LED

3.3 V logic | Editable schematic | Rev 1.1 | 2026-10-04



Grove cable: 1 yellow = BTN_N | 2 white = LED_CTL | 3 red = 3.3 V | 4 black = GND

Button is active LOW (pressed = LOW). LED_CTL HIGH lights D1; R6 100K keeps Q1 off while the MCU pin floats.

Q1 BSS138: pin 1 gate, pin 2 source, pin 3 drain. LED current is about $(3.3\text{ V} - 2.0\text{ V}) / 4\text{K7} = 0.28\text{ mA}$.

C3 0.1uF is the Grove port supply decoupling, as on the Maker Pi Grove ports. 3.3 V supply and signals only.

References: Maker Pi RP2040 sheets 2-3 (buttons, Grove ports, IO status LEDs); Maker Pi Pico sheets 1-2.

Values follow the Cytron references (10K / 1K / 0.1uF button, LED + 4K7 + BSS138) except R6: 100K instead of 3M3.

Grove pin order above is explicit (Cytron symbols number differently). SW2 uses switch pins 1 and 3; pads 2 and 4 unmapped - verify 1 and 3 are opposite contacts before wiring.